



Elecrow Crowpanel Advance 7 Enclosure



Alley Cat

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Summary

[ACOS] Case for the Elecrow Crowpanel Advance 7" ESP32 touchscreen tablet. Space for power switch and 8k lipo. UPDATED!

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Tags: [esp32](#) [tablet](#) [meshtastic](#) [alleycat](#) [elecrow](#)
[crowpanel](#) [acos](#)

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Suggested Print Settings:

Material: **PLA+** or **PETG**

Nozzle: **0.4mm**

Extrusion Width **0.4mm**

Layer Height: **0.2mm**

Perimeters: **3**

Infill: **20% Gyroid**

Top/Bottom Layers: **5**

Support Material: **None**

3D Printing Disclaimer:

Thank you for downloading my CAD design! I want to assure you that extensive testing and iteration went into creating this design. However, please keep in mind that 3D printing involves a wide range of variables, including different types of filament, printer settings, and calibration levels. These factors can significantly impact the final print.

Before concluding that the design is flawed, I encourage you to:

- **Check your printer calibration:** Ensure your printer is properly calibrated. **Calibrate filament shrinkage!**
- **Adjust print settings:** Try changing the print settings to achieve better results.
- **Resize the model:** Scaling the model up or down slightly may resolve fit issues.

Remember, these are development boards and sometimes have very fragile parts. If you encounter any issues or need assistance, please feel free to reach out to me. I am here to help and can provide guidance to achieve the best results. Claiming a design is bad without considering these variables does a disservice to the community and may discourage others from trying a design that could work perfectly with minor adjustments.

Thank you for your understanding and cooperation. Happy printing!

Description:

UPDATE 14FEB2026: Added the non SMA version.

Custom case for the [Elecrow CrowPanel Advance 7" HMI ESP32-S3 AI-Powered IPS Touch Screen](#). Requires 4x M3x8mm screw and 4x M3x12mm screws. Here are links to the parts I used:

Elecrow Crowpanel Advance 7" [Direct from Elecrow](#)

8000mAh MakerNova LiPo [Direct](#) / [Amazon](#)

17cm SMA Antenna [Direct](#) / [Amazon](#)

[Perfect Fit Folding Antenna](#)

Will not fit an antenna with bigger than 11mm base

[Right Angle SMA](#)

[Heat Set Inserts](#)

[Power Switch](#) or [Power Switch](#)

Assembly Video:

<https://www.youtube.com/watch?v=2JuRoskbhio>

Model files



acos-crowpanel-7-advance-sma-front-inserts.stl

☐ For Heat Set Inserts



acos-crowpanel-7-advance-front-inserts.stl

☐ Same but no SMA



acos-crowpanel-7-advance-sma-front.stl

☐ Screw threads into plastic



acos-crowpanel-7-advance-front.stl

☐ Same but no SMA



acos-crowpanel-7-advance-center-inserts.stl

☐ For Heat Set Inserts



acos-crowpanel-7-advance-center.stl

☐ Screw threads into plastic



acos-crowpanel-7-advance-buttons.stl

☐ Print two of these



acos-crowpanel-7-advance-sma-usb-rear.stl

☐ For RF expansion that has a USB



acos-crowpanel-7-advance-usb-rear.stl

☐ Same but no SMA



acos-crowpanel-7-advance-sma-rear.stl

☐ No USB on top



acos-crowpanel-7-advance-rear.stl

☐ Same but no SMA



acos-crowpanel-7-advance-complete.step

☐ STEP file so you can modify the design. CC BY-SA 4.0 still applies.

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